

3.3 Partial Derivatives

Motivation: Suppose we have a surface given by the two-variable function $z = f(x, y)$ and a point P (with coordinates of (a, b, c)) on the surface determined by f .

Suppose we want to know the rate at which the z -coordinate changes at P when we are moving across the surface parallel to the x -axis. This means that we would be moving along the curve C_1 toward P in the picture above. At this point P we can imagine a tangent line T_1 being constructed.

The slope of T_1 would then represent the rate at which the z -coordinate is changing at P when moving parallel to the x -axis.

Suppose we want to know the rate at which the z -coordinate changes at P when we are moving across the surface parallel to the y -axis. This means that we would be moving along the curve C_2 toward P in the picture above. At this point P we can imagine a tangent line T_2 being constructed.

The slope of T_2 would then represent the rate at which the z -coordinate is changing at P when moving parallel to the y -axis.

In Calculus I we found slopes of tangent lines to a curve $y = f(x)$ by calculating the derivative $f'(x)$ and evaluating it at the appropriate point.

In Calculus III for a function $z = f(x, y)$ we also need a way to find a derivative, but in a way that allows us to account for the direction that we are moving along the surface. This leads to the idea of what we call partial derivatives.

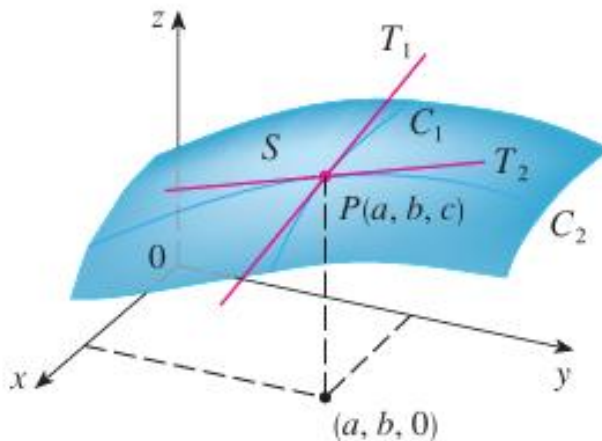
Note: The information that we are seeking above can be key when one wishes to better understand how quickly a surface is changing at a point.

Definition: If f is a function of two variables, its partial derivatives are the functions f_x and f_y defined by:

$$f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} \quad \text{and} \quad f_y(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h}.$$

Note: $f_x(x, y)$ represents the rate of change of z with respect to x when y is fixed, and $f_y(x, y)$ is the rate of change of z with respect to y when x is fixed.

Note: In the context of the motivation, we would use f_x to find the slope of the line tangent to the brown curve at P , and we would use f_y to find the slope of the line tangent to the green curve at P .



Notation: There are many notations for partial derivatives. We go over 7 possibilities. If $z = f(x, y)$, we write

$$f_x(x, y) = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} f(x, y) = \frac{\partial z}{\partial x} = f_1 = D_1 f = D_x f$$

$$f_y(x, y) = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y} f(x, y) = \frac{\partial z}{\partial y} = f_2 = D_2 f = D_y f$$

Example 1: If $f(x, y) = 2xy^3 + 6xy - x^7$, find $f_x(x, y)$ and $f_y(x, y)$.

According to the definition, to find $f_x(x, y)$, we think of y as a constant and differentiate with respect to x . So, $f_x(x, y) = 2y^3 + 6y - 7x^6$.

According to the definition, to find $f_y(x, y)$, we think of x as a constant and differentiate with respect to y . So, $f_y(x, y) = 6xy^2 + 6x$.

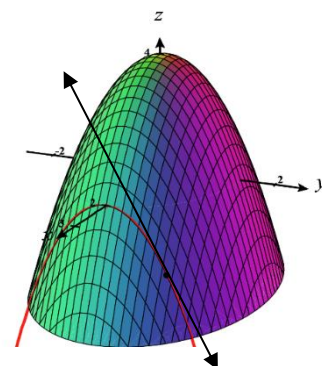
Example 2: Consider the surface given by $f(x, y) = 4 - x^2 - 2y^2$. Let C be the curve of intersection between this surface and the plane $x = 2$. Find the slope of the line tangent to C at the point $(2, 1, -2)$ on the surface.

We know that the answer to this question will be $f_y(2, 1)$. So, we note that

$$f_y(x, y) = -4y.$$

Then the answer is: $f_y(2, 1) = -4$.

Note: The instructor will illustrate the idea behind this example with a three-dimensional plotter.



Example 3: Find $\frac{\partial z}{\partial x}$ if z is defined implicitly as a function of x and y by $x^3 + y^3 + z^3 + 6xyz = 1$.

We differentiate both sides of the equation with respect to x while being careful to treat y as a constant and use the chain rule, we obtain:

$$3x^2 + 3z^2 \frac{\partial z}{\partial x} + 6yz + 6xy \frac{\partial z}{\partial x} = 0 \Leftrightarrow 3z^2 \frac{\partial z}{\partial x} + 6xy \frac{\partial z}{\partial x} = -3x^2 - 6yz$$

$$3 \frac{\partial z}{\partial x} (z^2 + 2xy) = -3(x^2 + 2yz) \Leftrightarrow \frac{\partial z}{\partial x} = \frac{-(x^2 + 2yz)}{z^2 + 2xy}.$$

Note: There is a fairly subtle point that could be made about the above example. In particular, we assumed the equation defines z implicitly as a function of x and y . What guarantees that the above equation defines z implicitly as a function of x and y ? In order to fully answer this question one needs the implicit function theorem (the proof of which is beyond the scope of this course).

We can easily extend the notion of partial derivatives to functions of more than two variables (though in these cases a geometric interpretation becomes more difficult). For example, Suppose that f is a function of three variables x , y , and z . Then, its partial derivative with respect to z is defined as

$f_z(x, y, z) = \lim_{h \rightarrow 0} \frac{f(x, y, z+h) - f(x, y, z)}{h}$. So, in order to compute f_z we think of x and y as constants and differentiate with respect to z .

Ex: If $f(x, y, z) = ye^{xy} \ln(z)$, then $f_z(x, y, z) = \frac{ye^{xy}}{z}$.

Definition: If f is a function of two variables, then f_x and f_y are functions of two variables. The functions $(f_x)_x$, $(f_x)_y$, $(f_y)_x$, and $(f_y)_y$ are called the second partial derivatives of f . If $z = f(x, y)$, we use the following notation:

$$(f_x)_x = f_{xx} = f_{11} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 z}{\partial x^2}$$

$$(f_x)_y = f_{xy} = f_{12} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 z}{\partial y \partial x}$$

$$(f_y)_x = f_{yx} = f_{21} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 z}{\partial x \partial y}$$

$$(f_y)_y = f_{yy} = f_{22} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 z}{\partial y^2}$$

Example 4: Find all four second partial derivatives of $f(x, y) = x^2 y^2 - y \sin(x^2)$.

We first note that $f_x(x, y) = 2xy^2 - 2xy \cos(x^2)$ and $f_y(x, y) = 2x^2 y - \sin(x^2)$.

This means that $f_{xx}(x, y) = 2y^2 - 2y \cos(x^2) + 4x^2 y \sin(x^2)$,

$$f_{yy}(x, y) = 2x^2, \quad f_{xy} = 4xy - 2x \cos(x^2), \text{ and}$$

$$f_{yx} = 4xy - 2x \cos(x^2).$$

Note: Interestingly, in the above example f_{xy} and f_{yx} are the same! The next theorem addresses this.

Note: As one would probably expect, the second partials f_{xx} and f_{yy} tell us something about the concavity of the curve of intersection between the surface given by $f(x, y)$ and a plane of the form $y = a$ or $x = b$. We will see some more applications of these second partials later in chapter 4.

Note: It is more difficult to perceive what the second partials f_{xy} and f_{yx} tell us. What f_{xy} does communicate is how the slope of the tangent line with respect to x changes as the value of y varies (similarly for f_{yx}). To see an animation of what we mean by this one can visit the following website:

https://web.archive.org/web/20180215172659/http://www.math.harvard.edu/archive/21a_fall_08/exhibits/fxy/index.html

Theorem: Suppose f is defined on a disk D that contains the point (a, b) . If the functions f_{xy} and f_{yx} are both continuous on D , then $f_{xy}(a, b) = f_{yx}(a, b)$.

Proof: Beyond the scope of this course

Note: In the most recent example, f_{xy} and f_{yx} are continuous on all of $\mathbb{R}^2$, and we saw that they were indeed equal.

We can have partial derivatives of order 3 and higher.

Example 5: If $f(x, y, z) = \sin(3x + yz)$, then find f_{xxy} .

We note that $f_x(x, y, z) = \cos(3x + yz)$. Then, $f_{xx}(x, y, z) = -\sin(3x + yz)$.

Finally, we have that $f_{xxy}(x, y, z) = -yz \cos(3x + yz)$.